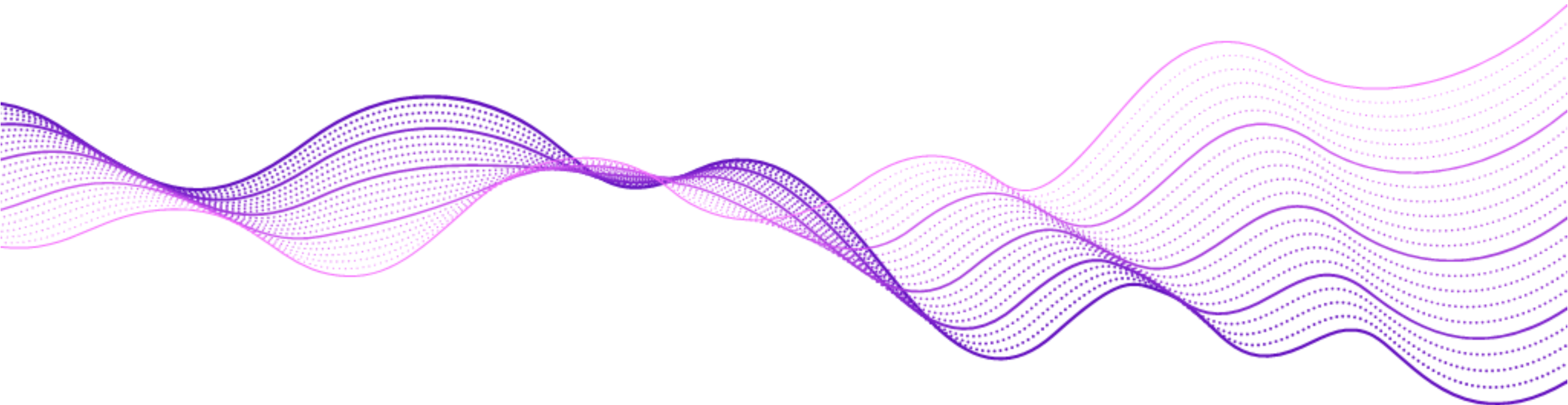


Amazon Bedrock AgentCore

Deploy, Operate & Scale Autonomous Agents on
AWS

Shivani.p.jha



AWS BEDROCK

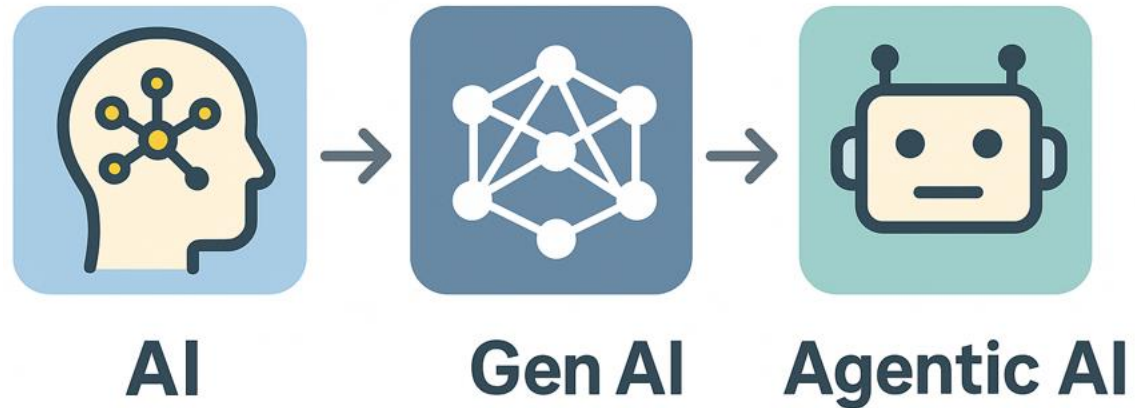
Agenda

- **Agentic AI Recap**
- **Overview of Bedrock**
- **Bedrock AgentCore Overview**
- **Core Functionality of Bedrock AgentCore**
- **Benefits of using Bedrock AgentCore**
- **Working with Agentcore**



Evolution of Agentic AI

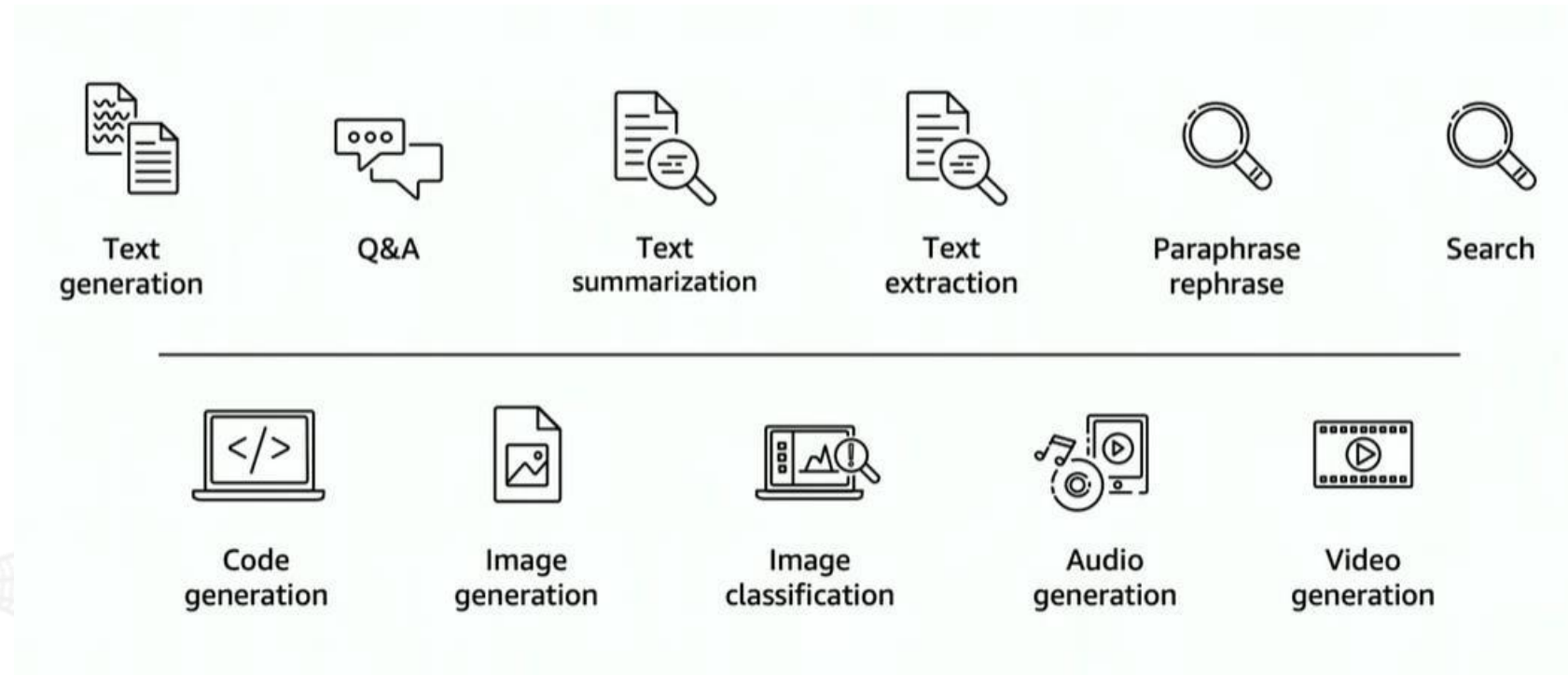
AI → ML → GenAI → Agentic AI



- **AI → ML:** From logic to learning
- **ML → Gen AI:** From prediction to creation
- **Gen AI → Agentic AI:** From creation to action

Gen AI Overview

- Generative AI is a type of artificial intelligence that can create new content and ideas, including conversations, stories, images, videos, and music.



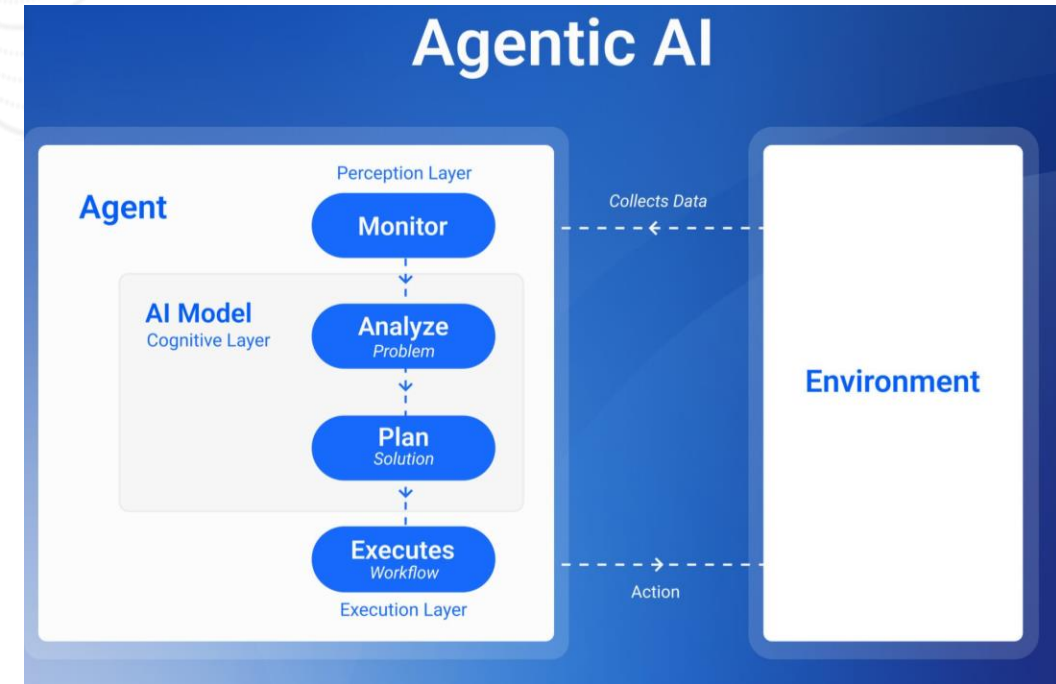
Agentic AI Overview

- AI systems that can Perceive → Plan → Act → Reflect toward a goal, beyond simple Q&A

Capabilities:

- Understand user intent
- Break down tasks into steps
- Use **tools/APIs/external data**
- Self-correct and continue until task is done
- **Difference vs. traditional AI:** not just responding — it **acts**.

To orchestrate workflows by reasoning, invoking APIs, and using tools with foundation models



Agentic AI Overview

Early Generative AI Phase: Used for simple content creation through prompts, producing outputs like paragraphs, images, or code. Characterized as singleton, reactive, and stateless.

Agent for Generative AI Phase: Leverages reasoning frameworks like ReAct, breaks complex tasks into simple steps, accesses external information, and integrates with APIs.

Multi-Agent Phase: Systems consisting of multiple fully autonomous agents that seamlessly execute multi-step workflows and adapt to changing environments.

AWS Bedrock Overview

Fast-track your generative AI applications and agents from prototype to production with confidence

fully managed service

industry leading foundation models (FMs) along with a broad set of capabilities that you need to build generative AI applications

simplifying development with security, privacy, and responsible AI

Experiment with a variety of top FMs, customize them privately with your data using techniques such as fine-tuning and retrieval-augmented generation (RAG)

create managed agents that execute complex business tasks

AWS Bedrock Overview

Available Models in Bedrock

Amazon
Anthropic
Cohere
DeepSeek
Luma AI
Meta
Mistral AI
OpenAI
....and more

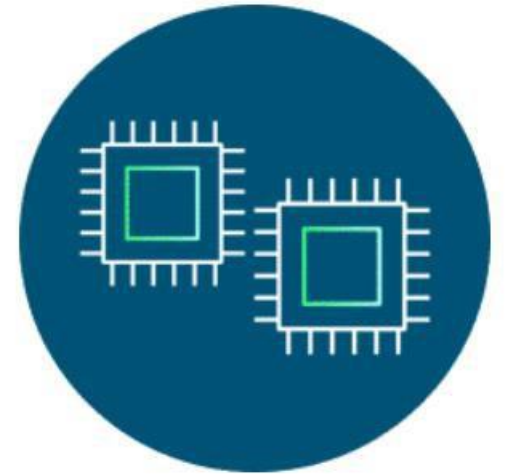
 Claude 3 Haiku Access granted Image to text, conversation, chat optimized Serverless	 Claude 3.5 Sonnet Available to request Legacy Image to text & code, multilingual conversation, complex reasoning &... Serverless	 Llama 4 Maverick... Available to request Fine Tuning, Text summarization, Code generation, Function calling, Advanced... Serverless	 Llama 4 Scout 17B... Available to request Fine Tuning, Text summarization, Code generation, Function calling, Advanced... Serverless
 Llama 3.3 70B... Access granted Tool Use, Code Generation, Advanced reasoning and decision making,... Serverless	 Llama 3.2 11B Visi... Available to request Chat optimized, Code generation, Complex reasoning analysis, Conversation,... Serverless	 Llama 3.2 1B Instruct Available to request Agents, Chat optimized, Code generation, Complex reasoning analysis, Conversatio... Serverless	 Llama 3.2 3B Instruct Available to request Agents, Chat optimized, Code generation, Complex reasoning analysis, Conversatio... Serverless
 Llama 3.2 90B Visi... Available to request Chat optimized, Code generation, Complex reasoning analysis, Conversation,... Serverless	 Llama 3.1 70B... Available to request Ability to follow instructions and tasks, improved reasoning and understanding ... Serverless	 Llama 3.1 8B Instruct Available to request Ability to follow instructions and tasks, improved reasoning and understanding ... Serverless	 Llama 3 70B Instruct Access granted Language modeling, Dialog systems, Code generation, Following instructions,... Serverless
 Llama 3 8B Instruct Available to request Text summarization, Text classification, Sentiment analysis Serverless	 Nova Premier Available to request Agents, Chat optimized, Code generation, Complex reasoning analysis, Conversatio... Serverless	 Nova Sonic Available to request Speech-to-speech, Speech-to-text, Realtime streaming speech recognition,... Serverless	 Nova Canvas Available to request Image-to-image Serverless

See here for supported foundation models from each provider:

<https://docs.aws.amazon.com/bedrock/latest/userguide/models-supported.html>

AWS Bedrock Overview

- AWS Inferentia and AWS Trainium are purpose-built ML accelerators that AWS designed from the ground up.
- The first-generation of AWS Inferentia delivers significant performance and cost-savings benefits for deploying smaller models.
- AWS Trainium and AWS Inferentia2 were built for training and deploying ultra-large generative AI models with hundreds of billions of parameters.



Bedrock Features

- Access to multiple foundation models via one API
- Custom model fine –tuning and Retrieval Augmented Generation(RAG)
- Its highly secured and maintains Data Privacy.
- Integrates with tools like Sage maker, Lambda, and Agentic AI (Agent Core)
- Use-cases:
 - Chatbots or Virtual Assistants
 - Text Summarization , Translation and Content Creation
 - Image Generation and analysis
- **Bedrock (general)** = the solid rock foundation of the Earth (LOL)
Amazon Bedrock (tech) = a foundation for building **AI applications safely and easily** using AWS

Why Agents and Agent Core

- What is an AI Agent ?
- Can you give us a simple example to understand AI agent?
- Features of AI agents:**
 - Decision making
 - Use tools like databases or web-browsers
 - Remember things
- ❑ **Why AgentCore** : Without a system like AgentCore, it's hard to run agents safely, securely, and reliably at scale.

Agentic AI Frameworks & Challenges



LlamaIndex

AutogenAI



Challenges

Complex orchestration of prompts & APIs

Context management (RAG, memory)

Security & compliance controls

Scaling & monitoring in production

Bedrock Agent Core

Brain and Control Center for your AI Agents

- Amazon Bedrock Agent Core is a **modular set of capabilities** designed to build, deploy and operate **production-grade agents** securely and at scale.
- It supports **any framework and model**, offering flexibility to integrate with your preferred tech stack.
- Agent Core capabilities are **optimized to work together or individually**, giving you the freedom to adopt just the pieces you need or the entire platform.
- Ideal for enterprise scenarios where you need **security, governance, scalability and customization**.

Agent Core empowers your enterprise to move from
experimental AI to **autonomous, mission-critical agents**.

Bedrock Agent Core

Ready-to-Use Agent Actions

Amazon Bedrock

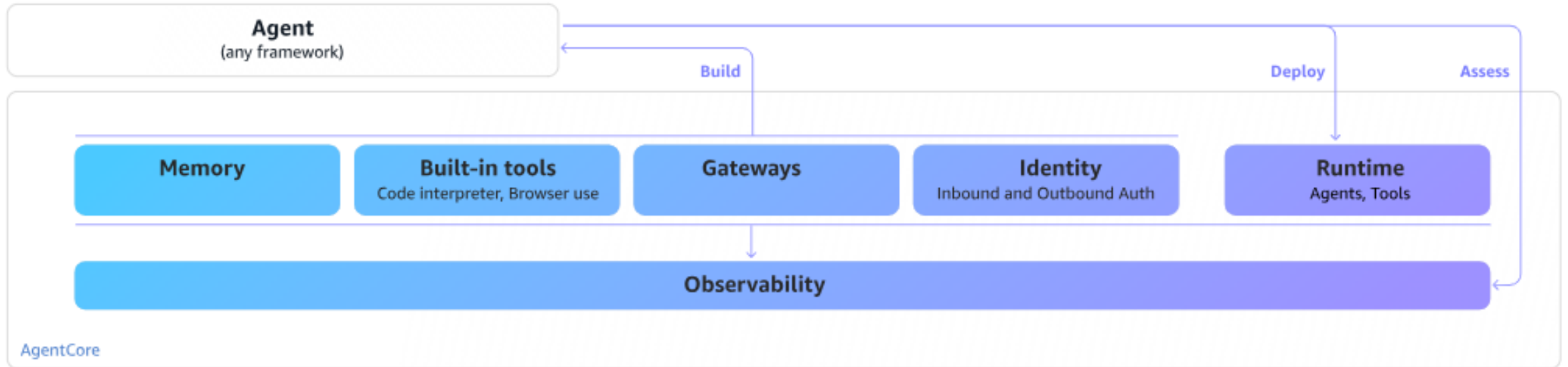
- Bedrock is the foundation: it provides access to high-performing foundation models (FMs) like Claude, Titan, Llama, and Mistral.
- It enables model orchestration, customization, fine-tuning, and RAG with Knowledge Bases.
- Ideal when the objective is to generate content or build AI-powered applications using LLMs.

Amazon Bedrock Agent Core

- Agent Core is the execution and operations layer for agents, not just models.
- It provides infrastructure for runtime, memory, identity, tool integration, and observability.
- Ideal when the objective is to build autonomous systems that can reason, act, call tools, and execute multi-step workflows safely in production.

Bedrock Agent Core

Ready-to-Use Agent Actions



Agent Core empowers your enterprise to move from **experimental AI** to **autonomous, mission-critical agents**.

Agent Runtime

Secure, Scalable Agent Execution

- Core execution environment for Bedrock Agents
- Manages sessions, state, scaling, and isolation
- Supports long-running, multi-step workflows
- Provides versioning and stable deployment endpoints
- Integrated monitoring and logging for traceability

Reliable **runtime orchestration** that lets agents reason and act end-to-end at production scale.

Built-in Tools

Ready-to-Use Agent Actions

- Pre-integareted capabilities that extend agent actions
- Code interpreter for secure data processing
- Browser interaction & web tool access
- Simplifies development and accelerates prototyping
- Works together with custom tool integrations

Enables **action-oriented behavior** beyond pure text generation.

Gateways

Secure Integration to External Systems

- Connect agents to APIs, Lambdas, and enterprise services
- Unified tool catalog with scalable request handling
- Automatic transformation from API to agent tool interface
- Enforces authentication and policy-controlled access
- Supports internal and third-party tools

A **single bridge** between your agents and operational systems.

Memory

Memory — Context and Continuity for Agents

- Short-term memory retains multi-turn conversational context
- Long-term memory captures insights across sessions
- No need for custom data stores or retrieval logic
- Fully governed access control and retention policies
- Improves personalization and decision accuracy

Agents that **remember**, leading to more useful and natural experiences.

Identity

Secure Access & Controlled Delegation

- Supports authentication via enterprise identity providers
- Agents assume least-privileged IAM roles for resource access
- Authorizes tool and data usage securely
- Full audit logs for compliance and accountability
- Critical for regulated and enterprise environments

Ensures agents act **securely** and with the **right permissions**.

Key Components of Bedrock AgentCore

1. Foundation Model

- Handles reasoning and decision-making
- Examples: Claude, Amazon Titan

2. Instructions (Agent Persona)

- Defines behavior, tone, boundaries
- “You are an IT support agent...”

3. Action Groups

- API integrations (Lambda-backed)
- Enable real-world actions (create ticket, fetch data, update system)

4. Knowledge Bases

RAG-powered enterprise context
Docs, PDFs, internal data

5. Orchestration Engine (AgentCore)

Manages planning
Chooses when to reason vs act
Controls execution flow

AgentCore Execution Flow (Agentic Loop)

1. User provides a goal
“Resolve VPN access issue”
2. AgentCore interprets intent
3. Plans steps
4. Invokes Action Groups
5. Observes responses
6. Refines plan
7. Completes goal

This is the **Think → Act → Observe → Repeat** loop.

Why Bedrock AgentCore is Enterprise-Ready

- **IAM-based security**
- **No training on customer data**
- **Fully managed execution**
- **Scalable and auditable**
- **AWS compliance & governance**

End-to-End Implementation Workflow

- **Choose the Model & Agent Code → to define intelligence and behavior**
Pick the LLM (reasoning) and your agent logic (what actions to take).
- **Create the Agent Runtime → to manage execution environment**
Provides session handling, scaling, isolation, and reliable workflow execution.
- **Define the Tools → to allow the agent to act**
Decide what external functions/services the agent is permitted to call.
- **Connect Tools via Gateway → to securely integrate APIs & services**
Gateway converts APIs/Lambdas into agent-accessible tools using MCP.
- **Add Memory → to remember context**
Short-term for multi-turn interactions; long-term for persistent knowledge.
- **Configure Identity → to control secure access**
IAM roles + authentication ensure the agent only accesses allowed data/tools.
- **Build & Test the Agent Workflow → to ensure real task completion**
Implement “understand → plan → call tools → validate → respond” and verify end-to-end behavior.

Benefits of using AgentCore

- **Simplicity and Speed:** Quick setup and deployment using AWS SDK or console.
- **Customization :** Add domain specific data ,customize behaviors ,workflows and response styles
- **Scalability & Reliability :** Fully managed by AWS ,scale agent to thousands of users effortlessly
- **Cost Efficiency :** Pay for usage ,no need for infrastructure maintenance.
-

“Instead of spending months coding and integrating AI systems manually, AgentCore lets you spin up a capable, business-ready AI agent in hours — integrated with your existing data and tools.”

Use cases:

Some possible use-cases where AgentCore can be very powerful:

Customer Support Agents: An autonomous agent that can handle helpdesk tasks, query internal systems, and respond to users.

Business Automation: Agents that call internal APIs (via Gateway) to automate workflows like lead generation, report generation, or ticketing.

Data Analysis: Use the code interpreter to let agents run Python/SQL to analyze data and give insights.

Web Automation: Agents that browse websites, fill out forms, scrape data, or perform web-based tasks using the browser tool.

Long-Term Agent Memory: Agents that “remember” past user interactions for personalization or continuing complex workflows.

Appendix

- [Amazon Bedrock AgentCore- AWS](#)
- <https://skillbuilder.aws/learn/PKCKV2ZP8A/aws-partner-amazon-bedrock-agentcore-overview/3XSV2HVVJS>
- <https://skillbuilder.aws/learn/UB7WCNMPGV/amazon-bedrock-agentcore-identity-tutorial/S29JUY1R7N>
- <https://www.bing.com/videos/riverview/relatedvideo?q=bedrock+agentcore+youtube&mid=D2A96921DB473D6EFAA3D2A96921DB473D6EFAA3&FORM=VIRE>

Q&A

Thank You